



PeerQA

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Introduction & Overview



The Problem

The development of automatic scientific QA systems is hindered by a lack of high-quality datasets, which are essential to help researchers navigate the overwhelming surge of academic literature.

Limitations of Prior Work:

Traditional scientific QA datasets often fail to reflect genuine inquiry because they rely on shallow annotations from non-experts rather than the deep domain knowledge required to generate realistic, high-quality questions.

The Solution

To address existing gaps, the authors introduced PeerQA, a real-world, document-level scientific dataset featuring authentic questions extracted from peer reviews and answers provided by the papers' original authors.

Core Tasks and Benchmarks

PeerQA serves as a benchmark for practical QA systems by evaluating three core tasks: evidence retrieval, unanswerable question classification, and free-form answer generation.

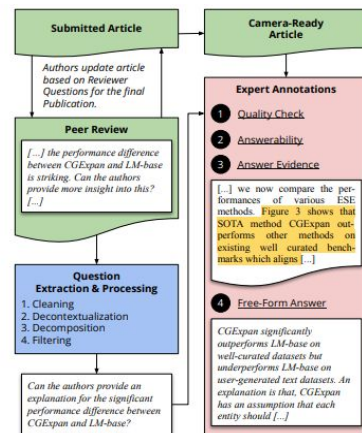
Methodology - Data Sourcing & Extraction

The authors pulled from a diverse set of venues, including major NLP conferences and Geoscience journals.

- Used GROBID to extract the full text and paragraphs, which also groups sentences into paragraphs
- Parsed peer reviews to extract 17,910 potential questions (identified by sentences ending in a question mark).
- These were often messy, highly contextual, and bundled together.

Initial Extraction: Decontextualization: Leveraged **InstructGPT** to rewrite messy or context-heavy review questions so they make sense as standalone queries.

Manual Filtering: Filtered the output to ensure questions were strictly information-seeking and focused on paper content (ignoring questions about typos or formatting).



Overview of the PeerQA data collection process

Methodology - Expert Annotation

The Annotators: The original authors of the 208 sourced papers.

Task 1: Evidence Retrieval: Authors highlighted the specific text segments within their papers that contained the answer.

Task 2: Answer Generation: Authors wrote free-form, expert-level answers based directly on the highlighted text.

Task 3: Answerability Flagging: Authors explicitly flagged questions that were impossible to answer given the information available in the final text.

Venue	Domain	Papers	Questions	Evidence	Free-Form	Ev. & FF.	Unanswerable
ICLR 23	ML	49	153	103	89	63	36
ICLR 22	ML	44	137	107	75	68	14
NeurIPS 22	ML	25	79	56	51	40	16
ARR 22	NLP	45	87	60	61	49	21
COLING 20	NLP	15	31	25	13	13	5
ACL 17	NLP	7	20	16	10	9	4
CoNLL 16	NLP	5	12	7	7	7	5
ESD 23	Geoscience	5	17	10	11	5	1
ESurf 23	Geoscience	3	16	16	9	9	0
F1000 22	Mixed	10	27	14	10	4	10
Total		208	579	414	336	267	112

Number of collected question-answer pairs per venue in PeerQA

Methodology - Baseline Experiments

The authors established **three baseline systems** to evaluate the dataset across three fundamental capabilities required for scientific QA:



Evidence Retrieval

Objective: Ranking document passages by relevance to a given question using various retrieval architectures (e.g., BM25, ColBERTv2, SPLADEv3).

Methodology: Evaluated paragraph vs. sentence granularity and tested decontextualization by prepending paper titles to snippets.

Metrics: Measured via Mean Reciprocal Rank (MRR) and Recall@10.

Answerability

Objective: A binary classification task determining if a question can be answered based on a provided context.

Context Variations: Tested across three settings: Gold (expert passages), RAG (top-k retrieved), and Full Text.

Evaluation: Benchmarked LLMs (e.g., Llama-3, GPT-4o) using Macro-F1 to manage class imbalance.

Answer Generation

Objective: Generating long-form responses based on the question and context (Gold, RAG, or Full Text).

Multi-Layered Evaluation: Utilized Rouge-L (lexical overlap), AlignScore (factual consistency), and Prometheus-2 (LLM-as-a-judge scoring).

Critique - What the Authors Did Well



Authentic "Expert" Ground Truth: Replaces superficial crowdsourced labels with authoritative annotations from the **original authors**, ensuring the highest possible reliability for evaluating model accuracy.

Real-World Scientific Inquiry: Features naturally occurring questions extracted directly from **actual peer reviews**, capturing the genuine concerns and complex information-seeking needs of domain experts.

Rigorous Long-Context Challenge: With documents averaging **~12,000 tokens**, the dataset serves as a "needle-in-a-haystack" benchmark, specifically testing a model's ability to synthesize evidence across non-consecutive sections.

Integrated Multi-Task Benchmarking: Provides a holistic framework to evaluate the entire QA pipeline—**retrieval, answerability classification, and generation**—allowing developers to pinpoint specific systemic bottlenecks.

Critique - What Could Have Been Better



Small Dataset Size: At only 579 QA pairs, it is quite small for fine-tuning modern, data-hungry models (a trade-off for using expert annotators).

Class Imbalance: Skewed heavily toward answerable questions (383) versus unanswerable (112), causing models to struggle with the minority class.

Extraction Technicalities: The PDF parser (GROBID) missed text, causing high-quality annotations to be discarded because they couldn't be mapped.

Language Limitation: The dataset is strictly English-only, limiting its global utility for multilingual scientific models.

Domain Limitation: The data is strictly from NLP conferences. The models would likely struggle to generate accurate rebuttals for other domains



Conclusion

Summary:

To conclude, PeerQA is a highly valuable, expert-verified dataset that introduces a very challenging task for long-context models. While it suffers from a small sample size and class imbalance, its authenticity makes it a strong benchmark for the NLP community. Thank you, and we will now take any questions.